Set & Map Interface

* set / Map
  - Duplicates are not allowed
  - Constant time search operation

0 1 2 3 4
[2 | 3 | 7 |]
list
→ O(N)

[7 | 3 | 2]
← set →

2 3 7

set.add(2)
set.add(3)
:
$\boxed{\text{set.contains(2)}}$
→ O(1)

Website
- Aditya ✓
- Rohit ✓

Set

email Id = [x@gmail.com,
y@gmail.com]

* Maps
  - (key, value)
  - Duplicate key are not allowed
  - Constant time search

map.containsKey(101)
→ O(1)

(Roll No, Name)
(101, Aditya)
(102, Rohit)
(103, Aditya)

103	102	101
Aditya	Rohit	Aditya

Map

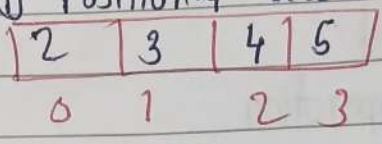
map.put(101, Aditya)
— (102, Rohit)
— (103, Aditya)

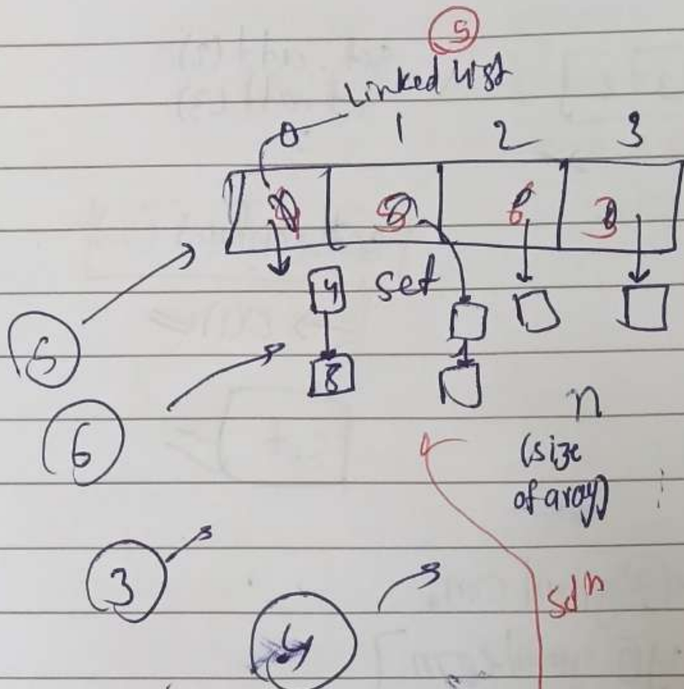
map.get(102); → O(1)

Set / Map

→ ① Positional access

list =





Bucket

$5 \% 4 = 1$

$\boxed{5 \% n}$

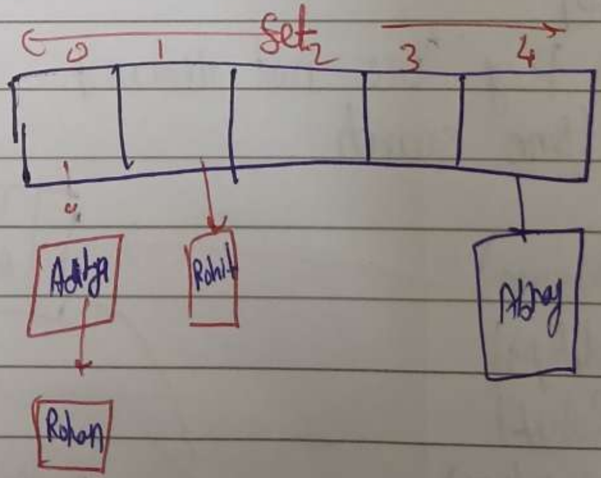
$6 \% 4 = 2$

$3 \% 4 = 3$

$4 \% 4 = 0$

$8 \% 4 = 0$

Collision problem



$n = 5$

Aditya, Rohit

hashCode()

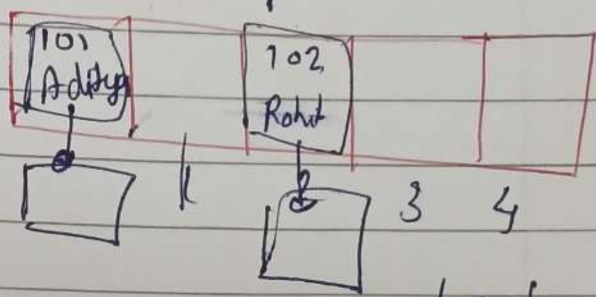
```
set.add ("Aditya")
   add ( T  value) {
   int i = value.hashCode();
       (i) % n
   }
```

Map
↳ (key, value)
 └ class

map.put(101, Aditya)
map.contains(101);
map.get(101);

```
class Node<K,V> {
    K key;
    V value;
}
```

Map




```
put(K, V) {
int i = key.hashcode();
(i % n)
}
```

due to collision problem we'll use Linked List in arrays

* Internals of Set/Map
Java → set (✗)
set ————→ Map ✓
↳ Duplicate (✗) ↳ Duplicate (✗)
↳ constant time (✓) ↳ constant (✓)